

**Source: same theorem  
for legs a b and  
hypotenuse c**

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graph TD; A[Source: same theorem for legs a b and hypotenuse c] --> B[Assumption: nonnegative lengths]; B --> C[Inference: encode triangle constraint algebraically]; C --> D[Inference: expand and simplify]; D --> E[Inference: isolate c squared]; E --> F[Conclusion: a squared plus b squared equals c squared];
```

**Assumption:  
nonnegative lengths**

**Inference: encode  
triangle constraint  
algebraically**

**Inference: expand and  
simplify**

**Inference: isolate c  
squared**

**Conclusion: a squared  
plus b squared equals  
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